

**Florida
Power**

CORPORATION
Crystal River Unit 3
Docket No. 50-302
Operating License No. DPR-72

February 27, 1993
3F0298-07

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

Subject: Request for NRC Approval of Topical Report - Boron Dilution by Reactor Coolant System Hot Leg Injection - Crystal River Unit 3

- References:
1. Topical Report BAW-10103A, Revision 3, July 1977, "ECCS Analysis of B&W's 177-FA Lowered-Loop NSS," Babcock & Wilcox
 2. NRC to B&W letter, Topical Report Evaluation, dated February 4, 1976
 3. FPC to NRC letter (3F0997-28) dated September 12, 1997, "Post-LOCA Boron Precipitation Mitigation Plan"
 4. FPC to NRC letter (3F1097-32) dated October 31, 1997, "License Amendment Request #223, Revision 0, Post-LOCA Boron Precipitation Mitigation Plan"
 5. FPC to NRC letter (3F0198-44) dated January 31, 1998, "Revision to Submittal Date for Hot Leg Injection via Reverse Flow Method of Post-LOCA Boron Precipitation Control"

Dear Sir:

Florida Power Corporation (FPC) is submitting a topical report for NRC review and approval, which describes an additional active method of preventing boron precipitation following a design basis loss of coolant accident (LOCA) at Crystal River Unit 3 (CR-3). This additional method, hot leg injection via reverse flow (HLI-RF) through portions of the Low Pressure Injection system (LPI), provides defense in depth for the current methods of post-LOCA boron dilution, Drop Line to Reactor Building Sump (DL-RB Sump) and Auxiliary Pressurizer Spray (APS). Submittal of this topical report describing the HLI-RF method by February 27, 1998, satisfies FPC's commitment made in References 3, 4 and 5.

The topical report is related to the subject of FPC's License Amendment Request (LAR) #223 (Reference 4). However, NRC review of the topical report is requested separate from review of LAR #223. Submittal of the HLI-RF method as a topical report separate from LAR #223 has been discussed with the NRC Staff.

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Florida Progress Company

The topical report consists of this submittal letter and the attached Description of Hot Leg Injection Via Reverse Flow Through Portions of the LPI System (Attachment A); FPC Calculation M97-0088 (Attachment B); letter from Ingersoll-Dresser Pump Company, dated November 12, 1997 (Attachment C); FPC Calculation M97-0098 with supporting Framatome Technologies Incorporated (FTI) Document 51-5000519-06 (Attachment D); and Acronyms and Abbreviations (Attachment E). FTI Document 51-5000519-06, provided as part of Attachment D, supersedes in its entirety FTI Document 51-5000519-02, submitted as Attachment D of LAR #223 (Reference 4).

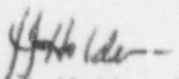
Hot leg injection via reverse flow (HLI-RF) was formally proposed to the NRC in Babcock and Wilcox (B&W) Topical Report BAW-10103 (Reference 1). That report described the basis for utilizing this boron precipitation mitigation option. The NRC evaluation of BAW-10103 (Reference 2) acknowledged the conceptual feasibility of this method of boron precipitation mitigation and indicated that its use would be reviewed by the NRC for individual plants on a case-by-case basis.

A summary of the passive and active methods of controlling post-LOCA boron concentration and preventing precipitation, as well as a detailed discussion of the HLI-RF method, is presented in Attachment A. A discussion of the mechanisms involved with boron precipitation following design basis LOCAs and the methods available at CR-3 have been previously provided to the NRC in License Amendment Requests #220 (approved as License Amendment #164) and #223.

FPC, with support from FTI, has evaluated the use of this method of post-LOCA boron precipitation control for CR-3 and has found it acceptable. However, implementation of the HLI-RF method at CR-3 requires NRC review and approval of the method. Therefore, FPC requests NRC review of the topical report and issuance of a safety evaluation report (SER). FPC will implement the HLI-RF method based on the SER when issued by the NRC. FPC will continue to participate with the B&W Owners Group in efforts to resolve this issue generically.

There are no commitments made in this submittal. If you have any questions regarding this submittal, please contact Ms. Sherry Bernhoft, Manager, Nuclear Licensing at (352) 563-4566.

Sincerely,


J. J. Holden, Director
Site Nuclear Operations

JJH:rer
Attachments

xc: Regional Administrator, Region II
NRR Project Manager
Senior Resident Inspector

ATTACHMENTS

- | | |
|--------------|---|
| Attachment A | Description of Hot Leg Injection Via Reverse Flow Through Portions of LPI System |
| Attachment B | FPC Calculation M97-0088, "Hydraulic Analysis for LPI Hotleg Injection to RCS" |
| Attachment C | Ingersoll-Dresser Pump Company letter to FPC dated November 12, 1997, "8HN-194 Decay Heat Pumps, Typical S/N - 1624920/21" |
| Attachment D | FPC Calculation M97-0098 R6, "Boron Dilution by Hot Leg Injection" (with supporting FTI Report, 51-5000519-06, "Boron Dilution by RCS Hot Leg Injection") |
| Attachment E | Acronyms and Abbreviations |

ATTACHMENT A

DESCRIPTION OF HOT LEG INJECTION VIA REVERSE FLOW THROUGH PORTIONS OF LPI SYSTEM

Introduction

A discussion of the mechanisms involved with boron precipitation following design basis loss of coolant accidents (LOCA) and the methods available for Crystal River Unit 3 (CR-3) was provided in License Amendment Requests (LAR) #220 and #223 (References 3 - 8). It was discussed in those submittals and associated references that certain LOCAs may result in the concentration of boric acid in the reactor core increasing to the point which it would potentially approach the solubility limit and begin precipitating out of solution. It was discussed in those submittals that passive and active methods would be needed to control the boron concentration. These methods will be initiated only if boron dilution is occurring in the sump or if sump sampling cannot be achieved. As described in those submittals, CR-3 uses the following methods to control post-LOCA boron concentration in the reactor core and prevent precipitation.

Passive Methods:

Reactor Vessel Vent Valve (RVVV) Overflow - This passive design feature results in natural circulation through the RVVVs and will automatically occur for cold leg breaks which result in sufficient refill of the reactor vessel (RV). Overflow through the RVVVs to the vessel downcomer replaces emergency core cooling system (ECCS) water of lower boron concentration from the lower plenum into the core region, which limits the maximum boron concentration to less than the solubility limit.

Hot Leg Nozzle Gap Flow - This passive design feature inherently utilizes natural circulation from the upper core region through the gaps between the hot leg outlet nozzles and core support shield to the downcomer. The flow of water through the hot leg nozzle gaps recirculates through the downcomer and mixing occurs thus diluting the concentration of boron in the core as the volume of water passed by the gaps is replaced with water of lower boron concentration.

Active Methods:

ECCS Flow Path Through the Reactor Vessel - This method applies to those breaks on the hot leg side of the RV. The ECCS flow path through the reactor coolant system (RCS) is from the cold leg, into the downcomer, through the core, and out the hot leg (to the break) which will supply a continuous flow of ECCS water through the RV, limiting the boron concentration to less than the solubility limit. No additional operator actions are required to ensure dilution. Breaks on the cold leg side of the RV do not provide this flow path.

Drop Line to the Reactor Building (RB) Sump (DL-RB Sump) - In this method, aligning the decay heat drop line to the RB sump will allow gravity feed of reactor coolant from the hot leg to the RB sump and recirculation back to the RV through the ECCS. Figure 1 shows the system flow paths when using the DL-RB Sump method. This will result in injection flow through the core at a rate equal to the drop line flow rate plus core boil-off.

There are precautions with this method regarding when the drop line to the sump can be opened. Limitations associated with this method are currently prescribed in CR-3 Emergency Plan Implementing Procedure EM-225B, "Post-Accident Boron Concentration Management."

Auxiliary Pressurizer Spray (APS) - In this method, water from the RB sump is injected into the pressurizer via the pressurizer spray. This water then flows into the hot leg forcing a reverse flow through the reactor core. This method provides a constant flow of water with lower boron concentration through the core, diluting the RCS boron concentration in the reactor core region, thereby preventing boron precipitation. The flow path is through the APS line, supplied from the low pressure injection (LPI) system, out of the pressurizer through the surge line into the hot leg, and then into the RV. Figure 2 shows the system flow paths when using the APS method. APS is effective when the spray flow rate exceeds the boil-off rate.

Proposed New Method

In addition to the above methods, FPC has evaluated another active method that would be available for use at CR-3. This method provides hot leg injection via reverse flow (HLI-RF) through portions of the LPI system. This method involves aligning the ECCS flow path in a manner that has not been previously approved by the NRC for CR-3. This alignment involves securing one train of ECCS (if running), aligning the system to provide reverse flow through the idle LPI pump, into and through the decay heat drop line to the hot leg, thereby providing reverse flow through the reactor core.

As part of the original licensing basis for boron precipitation control at CR-3, Babcock and Wilcox (B&W) Topical Report BAW-10103 (Reference 1) discussed the alignment of a part of the LPI injection to flow backward through the drop line into the hot leg. The NRC evaluation of the topical report acknowledged the conceptual feasibility of this method of boron dilution and indicated its application would be reviewed on a case-by-case basis for individual plants.

FPC, with support from Framatome Technologies Incorporated (FTI), evaluated the use of this method of post-LOCA boron precipitation control for CR-3, and has found it acceptable. This method will provide an additional means of post-LOCA boron precipitation control. The results of this evaluation are included below. The FTI evaluation of this method is documented in FTI Report 51-5000519-06, provided as Attachment D. Additional discussion of HLI-RF, including a demonstration of its effectiveness, can be found in Reference 9.

Description of HLI-RF Method

Figure 3 shows the system flow paths when using the HLI-RF method. In this method of post-LOCA boron dilution, ECCS flow is simultaneously injected into the RCS via the following three flow paths:

- LPI through the core flood line nozzle to the reactor vessel downcomer
- High pressure injection (HPI) system to the RCS cold legs, with water supply provided by the operating LPI pump
- Reverse flow through the decay heat drop line to the RCS hot leg

The ECCS flow rate will be established through each of these injection paths to ensure sufficient flow to the core for removal of decay heat and maintain adequate core cooling. To accomplish this during the time when HLI-RF may be needed to prevent post-LOCA boron precipitation, the flow rate through the flow paths will exceed: 1) the core boil-off rate necessary to remove decay heat and 2) hot leg nozzle gap flow.

To provide the ECCS injection into the RCS as described above, one operating LPI pump continues to be aligned to provide flow through one core flood line nozzle and to the suction of one HPI pump which is also providing ECCS injection flow. After ensuring adequate operation of one ECCS train, the opposite ECCS train is secured. The LPI cross connect line is opened and a portion of the ECCS flow is routed backwards through the idle LPI pump, through the decay heat drop line, and into the hot leg. This alignment reverses the typical flow direction in the idle LPI pump and decay heat drop line.

The flow path inside the reactor vessel during HLI-RF before and after core boiling suppression is shown in FTI-51-5000519-06, Figures 3 and 4, respectively (Attachment D). Flow from the hot leg enters the reactor vessel outlet plenum where a portion of the hot leg injection may flow through the hot leg nozzle gap to the downcomer. The remaining HLI-RF flows into the upper plenum and to the core. As discussed in the following section, the minimum required HLI-RF flow rate will exceed the core boil-off rate plus the flow that could potentially bypass the core by flowing through the hot leg nozzle gap. A net reverse flow of lower boron concentration fluid through the core will arrest and retard the process of concentrating of boron which will prevent reaching the solubility limit. The net reverse flow through the core will then pass through the lower plenum and downcomer to the cold leg where it will flow out the cold leg break. The analyses to support this method are discussed below.

ECCS Requirements for HLI-RF

Preventing post-LOCA boron precipitation using HLI-RF will involve a line-up of the ECCS flow path that has not been previously approved by the NRC for CR-3. This line-up includes securing one train of ECCS (if running), providing reverse flow through the idle LPI pump and the decay heat drop line, and providing reverse flow through the reactor core.

In FTI Report 51-5000519-06, FTI recommends maintaining flow rates through the RCS injection paths as described in Table 1. These flow rates provide both core cooling and boron dilution.

Table 1
Flow Rates for HLI-RF

Flow Path	Flow Rate (gpm)
LPI flow through core flood line nozzle to the reactor vessel downcomer	approximately 1000
HPI system to the RCS cold leg	600
Reverse flow through the decay heat drop line to the RCS hot leg	500

An example provided in FTI Report 51-5000519-06 states that at an RCS pressure of 72 psia (corresponding to a saturation temperature of 305° F), all of the boron initially injected into the RCS from the Core Flood Tanks (CFT) and Borated Water Storage Tank (BWST) will not result in boron precipitation. At higher temperatures and pressures, the boron solubility limit is greater than the maximum possible core boron concentration. In addition, based on 1.0 times the ANS 1971 fission product decay heat, the earliest time following a LOCA at which boron concentration in the reactor core could reach the solubility limit is approximately 13 hours. Taking into consideration the Appendix K decay heat level of 1.2 times ANS 1971, the time to reach the solubility limit is approximately 7.5 hours. This is based on a large break LOCA (LBLOCA), where the RCS pressure and temperature rapidly approach RB conditions. In the case of small break LOCAs (SBLOCA), the RCS pressure and temperature can remain elevated above the containment pressure. The higher saturation temperature results in a higher boron solubility limit that cannot be reached until later in the transient.

The HLI-RF flow rate was calculated to ensure a net reverse flow through the core that will dilute the core boron concentration, considering the core boil-off rate and an allowance for flow that could potentially bypass the core by flowing through the hot leg nozzle gap. In FTI Report 51-5000519-06, FTI recommends that the HLI-RF alignment provide up to 600 gpm flow for one HPI pump, a minimum hot leg injection flow of 500 gpm for boron dilution, and approximately 1000 gpm into one core flood line nozzle. Following either a SBLOCA or LBLOCA, a minimum HLI-RF flow of 500 gpm will provide sufficient reverse flow to dilute the core boron concentration and prevent boron precipitation at 5 hours and later. Figures 5 through 8 of FTI-51-5000519-06 (Attachment D) reflect boron concentrations versus time for LBLOCA and SBLOCA based on use of HLI-RF.

As decay heat decreases with time, the minimum required HLI-RF flow will eventually be sufficient to suppress core boiling. At the point that the HLI-RF flow is sufficient to overcome core boiling, boron precipitation will no longer be a concern. Re-establishment of subcooling and suppression of core boiling eliminates the mechanism that concentrates the boron, thereby eliminating the boron concentration concerns for the remainder of the event.

System Analysis

LPI Flow Rate

FPC performed a hydraulic analysis of the LPI system to demonstrate that the HLI-RF flow requirements determined by FTI would be satisfied. This analysis also evaluated the impact of reverse flow through the affected LPI components, including the LPI pump. This analysis, FPC Calculation M97-0088, is provided as Attachment B.

Hydraulic modeling of the LPI system concluded that the operating LPI pump is capable of supplying 500 to 1000 gpm of reverse flow through the drop line throughout the range of RCS pressures for which reverse flow would be required, while simultaneously providing necessary ECCS flow. At higher pressures when 1000 gpm flow through the drop line could not be achieved, flow through the normal LPI injection path may be throttled in order to maintain reverse flow through the drop line greater than 500 gpm. The results of the analysis are summarized in Table 2.

Table 2
HLI-RF Flow Rate

RCS Pressure (psig)	Drop line Reverse Flow (gpm)	Normal LPI Flow (gpm)	LPI Pump Flow (gpm)
0	999.8	1350.1	3048.3
10	999.9	1350.1	3048.5
20	1000.4	1350.0	3048.8
30	1000.2	1349.2	3047.9
50	1000.9	1350.1	3049.6
65	1000.3	1349.9	3048.6
70	1000.1	1349.6	3048.1
80	990.6	1350.5	3039.5
85	834.8	1525.2	3059.6
90	698.3	1650.0	3048.0
95	600.7	1700.6	2999.9
100	595.8	1575.1	2872.0
110	596.9	1275.3	2575.4
120	583.8	1000.3	2289.3

NPSH Evaluation

Net positive suction head (NPSH) required and available for the LPI pump in this configuration was also evaluated to ensure cavitation within the pump would not occur. The analysis concluded that adequate NPSH is available for the LPI pumps. The analysis performed to support this conclusion, FPC Calculation M97-0088, is provided as Attachment B.

Component Evaluation

Components within the flow path for this method (HLI-RF) were evaluated for the effects of reverse flow. The critical components evaluated were:

- the windmilling LPI pump, effects upon seals, motor, and potential pressure drops
- the control valves which will be used to control flow in the reverse direction
- the heat exchangers

Effects upon the LPI pump seals in backflow conditions have been evaluated by the pump seal manufacturer. This evaluation concludes that the LPI pump seals will perform acceptably under the approximate range of 500 to 1000 gpm backflow conditions. The letter from the pump seal manufacturer, John Crane Incorporated dated August 13, 1996, is included in FPC Calculation M97-0088 (Attachment B).

The LPI pump manufacturer, Ingersoll-Dresser Pump Company, has reviewed the pump with regard to running in reverse rotation. The conclusion was that the pump will not have a problem running up to full speed in reverse, with no damage or loss of service life. The letter from Ingersoll-Dresser Pump Company, dated November 12, 1997, is provided as Attachment C.

The differential pressure drop across the windmilling decay heat pump has been determined and was used in the development of the hydraulic models previously described. Windmilling an AC motor at speeds of up to 890 rpm (50% of 1780 rpm) does not create problems for the motor or bearings. When HLI-RF is secured, pump coastdown from reverse rotation will occur prior to restarting the idle LPI pump in the forward flow direction.

The ability of the control valves DHV-110 and DHV-111 in the LPI pump discharge to regulate flow in the approximate range of 500 to 1000 gpm was discussed with Crane Valve, the supplier of these valves. Crane Valve confirmed that these valves would perform acceptably for an extended period. The record of this communication with Crane Valve is included in FPC Calculation M97-0088 (Attachment B).

LPI injection flow will be throttled to approximately 1000 gpm by DHV-5 and DHV-6. These valves have been evaluated for similar throttling applications. Based on correspondence from Crane Valves and communications with FPC's Material Technology group, structural integrity of the valves is expected to be maintained in this configuration.

Hydraulically, the heat exchangers are unaffected by the direction of flow on the tube side. The heat transfer characteristics of the heat exchangers would be affected in that the amount of heat transfer through this heat exchanger is less than the amount when tube flow is in the normal direction since these heat exchangers are counterflow heat exchangers. Heat transfer to the decay heat closed cycle cooling water system is, therefore, bounded by existing heat transfer analysis. No credit is taken for any heat transfer through the reverse flowing heat exchanger.

This review of components subjected to reverse flow conditions indicates that all components will perform acceptably under reverse flow conditions of 500 - 1000 gpm.

FTI has also evaluated the impact of the hot leg injection on the hot leg piping, the reactor vessel, and the reactor vessel internals. The evaluation includes:

- Steam water interaction and/or cold water-hot water interactions (water hammer)
- Stratified liquid flow in the hot leg pipe
- Temperature rates of change for the reactor vessel and the reactor vessel internals
- Operation within the Pressure-Temperature limits
- Reverse flow in the core

The results of the evaluation are documented in FTI Report 51-5000519-06, provided in Attachment D. The evaluation indicates that there are no thermal-hydraulic, mechanical, or core conditions that will prevent the use of HLI-RF to dilute core boron concentration.

Procedures

Guidance on the use of the HLI-RF method is provided in Emergency Plan Implementing Procedure EM-225B, "Post-Accident Boron Concentration Management." A note in that procedure states that this method cannot be implemented until after appropriate approval by the NRC of this method. Following NRC review and approval, this post-LOCA boron dilution method will be implemented only at the direction of the Technical Support Center (TSC) similar to the other active boron dilution methods.

Notwithstanding the note in EM-225B, FPC's position is that, pursuant to 10 CFR 50.54(x) and (y), this method could be used at CR-3 prior to NRC review and approval, but only as an emergency action and only if the other available methods of boron dilution are not adequately controlling boron concentration in the reactor vessel. Approval by the accident assessment team assembled in the TSC and the Emergency Coordinator would be required prior to use of this method under these circumstances.

FPC considers that providing this guidance and controls on use of HLI-RF in EM-225B at this time helps assure nuclear safety since it ensures that all options for controlling boron concentration will be readily available during post-accident conditions.

Conclusion

FPC, with support from FTI, has evaluated the use of the HLI-RF method of post-LOCA boron concentration control for CR-3 and has found it acceptable. Evaluation of this method included determination of ECCS flow rates for core decay heat removal and boron dilution, hydraulic analysis of the LPI system, and review of components for reverse flow effects. This is an additional method for post-LOCA boron concentration control.

References

1. Topical Report BAW-10103A, Revision 3, July 1977, "ECCS Analysis of B&W's 177-FA Lowered-Loop NSS," Babcock & Wilcox
2. NRC to B&W letter, Topical Report Evaluation, dated February 4, 1976
3. FPC to NRC letter, 3F1097-32, dated October 31, 1997, "License Amendment Request #223, Revision 0; Post-LOCA Boron Precipitation Prevention"
4. FPC to NRC letter, 3F1097-08, dated October 31, 1997, "License Amendment Request #220, Revision 1; Revision of Operating License Condition 2.C.(5); (TAC No. 99128)"
5. FPC to NRC letter, 3F1297-21, dated December 1, 1997, "Clarification to License Amendment Request #220, Revision 1 (TAC No. 99128)"
6. FPC to NRC letter, 3F1297-12, dated December 4, 1997, "Additional Information Regarding the Post-LOCA Boron Precipitation Prevention Plan for CR-3"
7. FPC to NRC letter, 3F1297-43, dated December 13, 1997, "Additional Information on License Amendment Request #220 - Revision of Operating License Condition 2.C.(5) (TAC No. M99128)"
8. FPC to NRC letter, 3F1297-44, dated December 20, 1997, "Additional Information Regarding License Amendment Requests #220 & #223 (TAC Nos. M99128 & M99892)"
9. Framatome Technologies Inc. Document 86-1266272-01, "Post-LOCA Boron Concentration Management for CR-3," February, 1998

Figure 1 Drop Line to Reactor Building Sump (Flow Schematic)
(Flow path is indicated by dashed line)

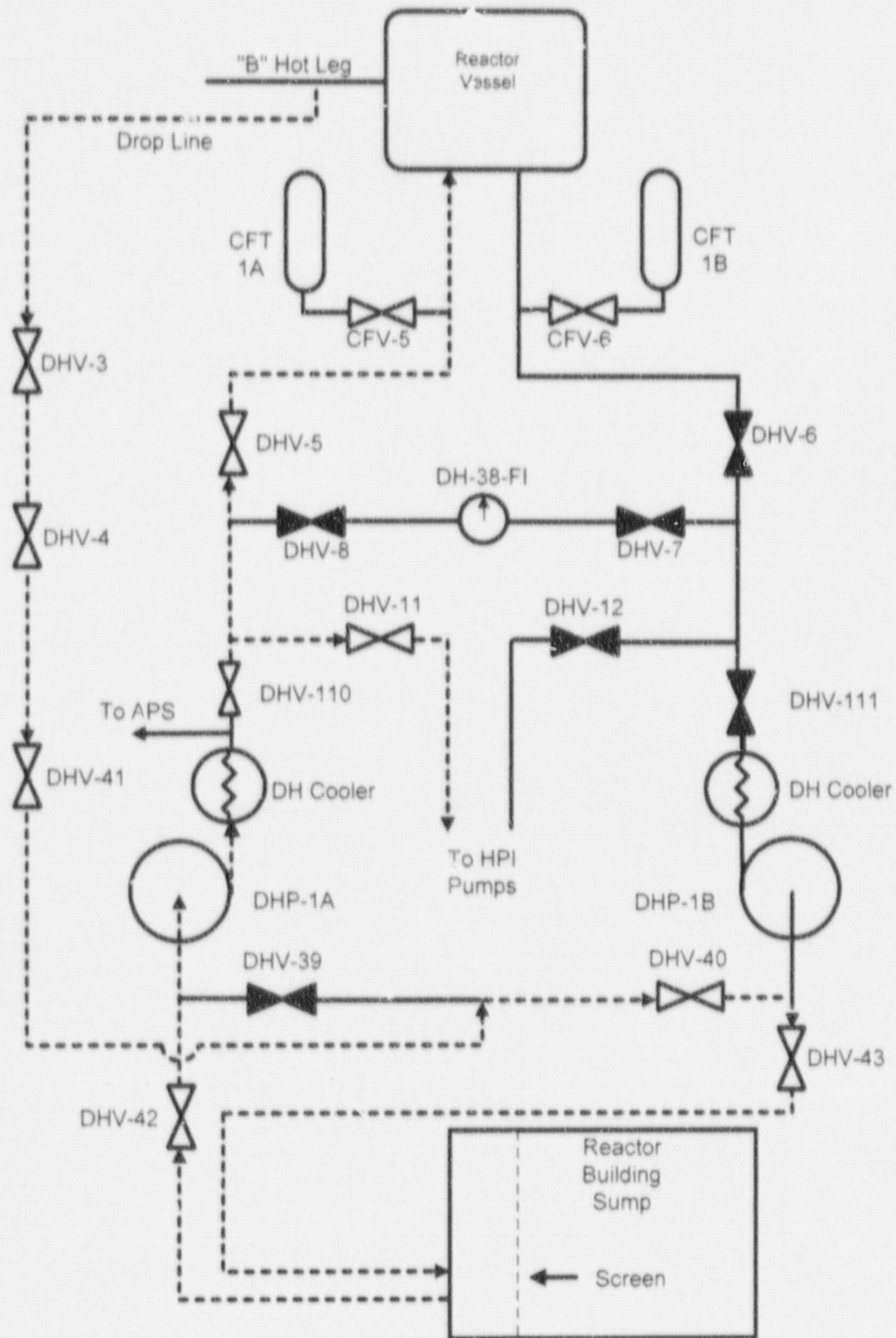


Figure 2 Auxiliary Pressurizer Spray (Flow Schematic)
(Flow path is indicated by dashed line)

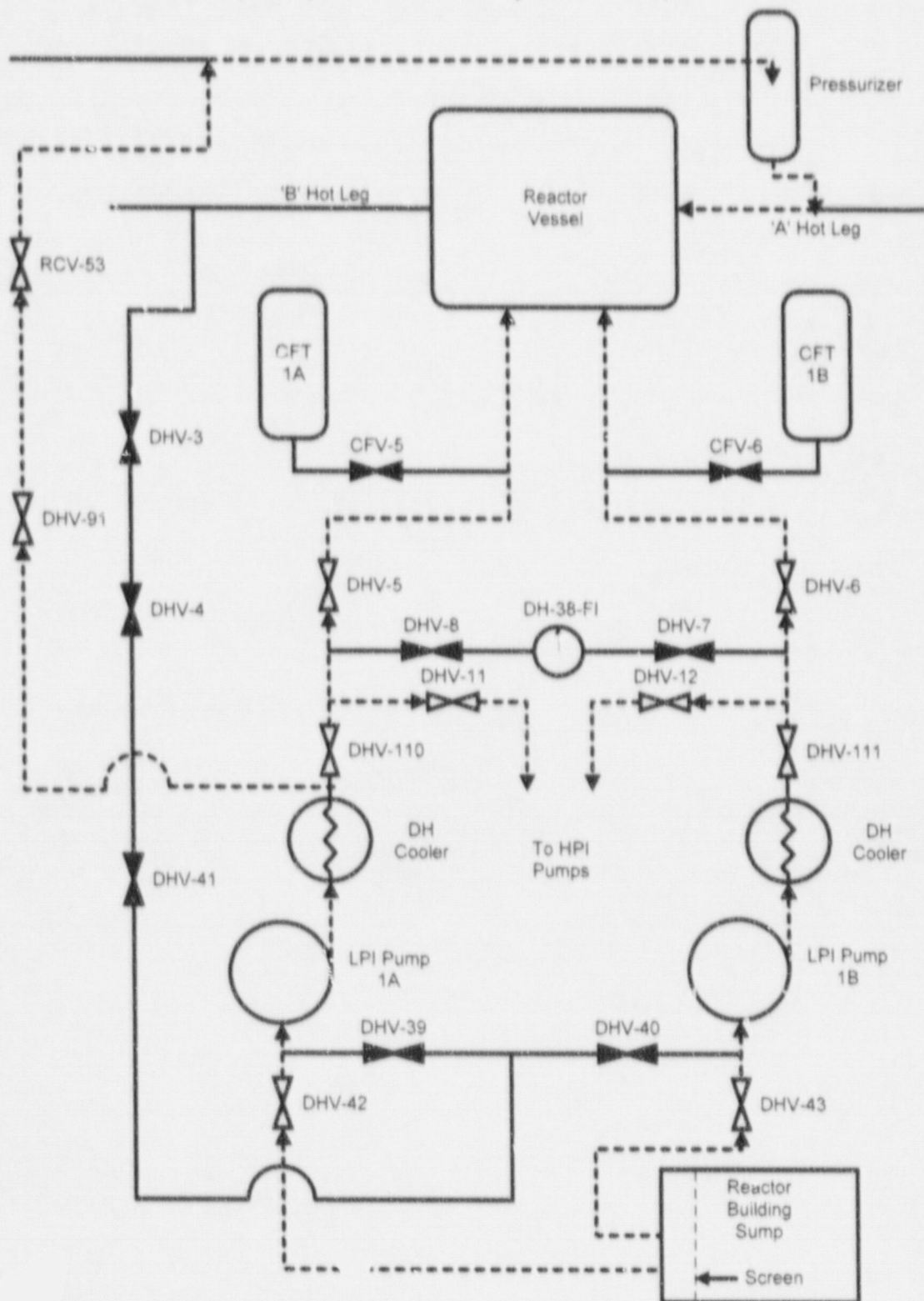
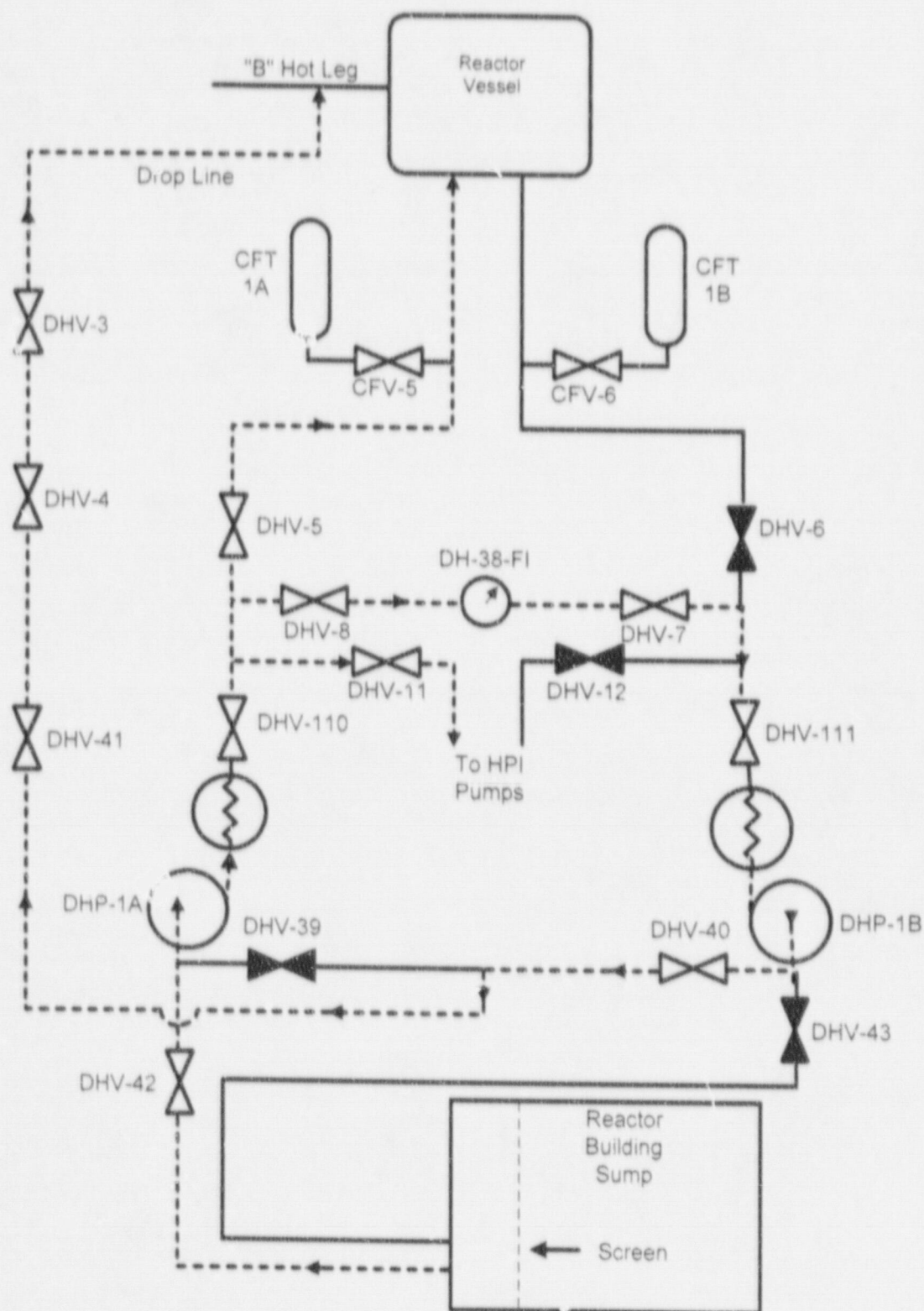


Figure 3 Hot Leg Injection via Reverse Flow (Flow Schematic)
(Flow path is indicated by dashed line)



ATTACHMENT E

ACRONYMS AND ABBREVIATIONS

APS	Auxiliary Pressurizer Spray
B&W	Babcock and Wilcox
BWST	Borated Water Storage Tank
CFT	Core Flood Tank
CR-3	Crystal River Unit 3
DL-RB Sump	Drop Line to Reactor Building Sump (mitigation method)
ECCS	Emergency Core Cooling System
FPC	Florida Power Corporation
FTI	Framatome Technologies Incorporated
gpm	gallons per minute
HLI-RF	Hot leg injection via Reverse Flow (mitigation method)
HPI	High Pressure Injection system
LAR	License Amendment Request
LBLOCA	Large Break LOCA
LOCA	Loss of Coolant Accident
LPI	Low Pressure Injection system
NRC	U.S. Nuclear Regulatory Commission
ppm	parts per million
RB	Reactor Building
RCS	Reactor Coolant System
RV	Reactor Vessel
RVVV	Reactor Vessel Vent Valve
SBLOCA	Small Break LOCA
TSC	Technical Support Center

ATTACHMENT B
TO 3F0298-97

FPC CALCULATION M97-0088, Revision 0
Hydraulic Analysis for LPI Hotleg Injection to RCS
(Computer Output omitted)